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CIN : U62099RJ2025OPC100267

Department Wise Audit Report

Research & Development

Field Trials, Grants & Technical Reporting — Ritu Panwar

An independent operational review of the R&D department, evaluating trial data fragmentation, key-person bottlenecks, and technical documentation workflows, prepared as part of the EF Polymer Private Limited organization-wide operational audit by Creative Upaay (brand owned by Cinnovate Solutions (OPC) Private Limited).

CLIENT	EF Polymer Private Limited
PREPARED BY	Creative Upaay
DOCUMENT VERSION	1.0
DATE ISSUED	18th July 2026
REVIEW SCOPE	R&D, National and International Field Trials & Product Validation



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SECTION 01 — EXECUTIVE SUMMARY

Department Overview

Scope, current state, and headline findings

The R&D team manages new product development, crop validation trials, regulatory registration, and international technical documentation. Operating across India, the US, EU, and Japan, the team forms the scientific backbone of EF Polymer Private Limited.

The audit identified a severe key-person dependency on the Product Development Manager (Ritu Panwar), who single-handedly manages multi-stage protocol approvals and grant drafting for all regions. Additionally, research findings are fragmented across 8–10 independent Excel spreadsheets. This lack of a unified database leads to redundant testing and risks institutional knowledge loss.

18–20 ACTIVE TRIALS AND AUDITS	8–10 EXCEL TRACKERS	3 APPROVAL STAGES	1.5 / 5.0 OVERALL MATURITY SCORE
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About the Maturity Score

A 1–5 scale measuring how systematised, documented, and independently operable the department is. Scored across five dimensions: Process, Technology, Data Quality, Documentation & Knowledge Management.

1 — Ad hoc ————— 5 — Optimised

■ **1.5** Current state ■ **3.2** Post-roadmap target ○ **5.0** Best-in-class benchmark



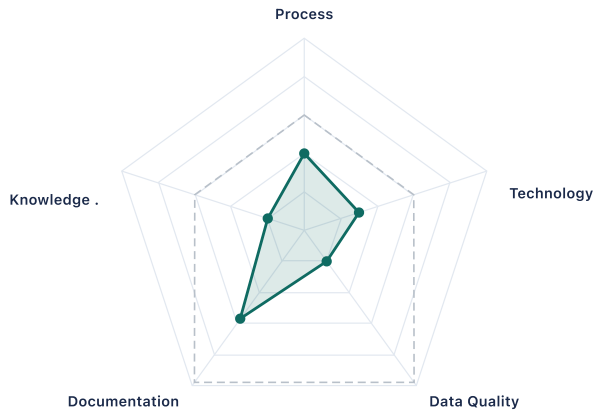
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SECTION 02 — BENCHMARKING

Department Maturity Assessment



Solid line: current state. Dashed line: target state (post-roadmap).

R&D scores 1.5 overall — The department is a critical knowledge center, yet scores only 1.0 in Knowledge Management and Data Quality due to over-reliance on individual memory and 8-10 disparate Excel trackers.

Migrating to a unified database and formally delegating approvals will structurally stabilize the department and preserve institutional IP.

DIMENSION	CURRENT	TARGET	GAP
Process Standardization	2.0	3.0	1.0
Technology Utilization	1.5	3.0	1.5
Data Quality & Accessibility	1.0	3.5	2.5
Documentation	2.0	3.5	1.5
Knowledge Management	1.0	3.0	2.0
Overall	1.5	3.2	1.7



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Team Structure

Reporting hierarchy within R&D, led by Ritu Panwar (Product Development Manager) under Rishabh Doshi (Production Manager) and Narayan Lal Gurjar (Founder & CEO).



Tools in Use



Google Sheets

Project tracking — 8–10 simultaneous trackers across India, Japan, EU & USA trials



Google Docs

Scientific white papers, technical reports, registration & compliance documentation



Google Slides

Presentations for B2B pitching, technical training & partner briefings



Google Drive

Central file upload, folder management & cross-team document sharing



Power BI

Active onboarding — GIS dashboards & centralized R&D data reporting (Shubham)



Paper AI

Online scientific paper referencing tool for literature reviews & proposal back-linking



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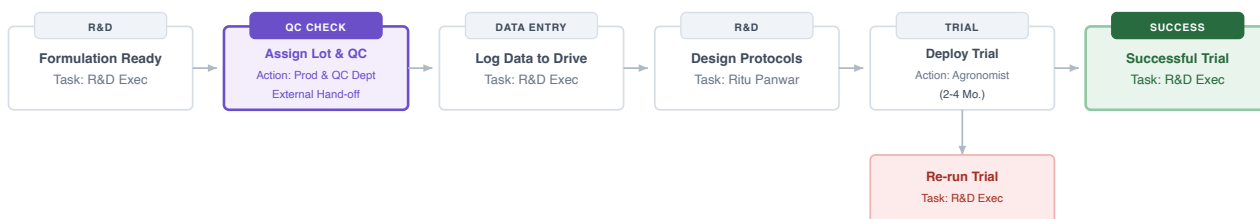
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SECTION 03 — PROCESS DETAIL

Current Workflows

Where each finding sits inside the operational process

Product Validation & Trial Lifecycle



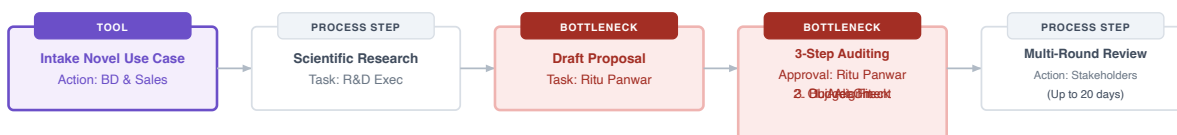
What is good: The sequence from formulation to field deployment is logically ordered and well-defined across teams.

What needs to be fixed: Validation is R&D's largest cost center. Untracked findings lead to repeated trials and inflated validation budgets. Data is logged manually into scattered sheets.

Bottlenecks: Retrieving historical data from unlinked Google Drive trackers risks data loss and prevents learning from past trials.

Dependencies: 3-department handoff — Production assigns the lot number → QC runs the 24-hr test and issues a QC report → R&D logs the result. Any break here derails the trial.

B2B Proposal & Protocol Drafting — P-01



What is good: Comprehensive 3-step technical and budget screening prevents unviable trials from reaching the field.

What needs to be fixed: All international proposals and scientific grants bottleneck on a single person (PD Manager). Requires 2-3 manual drafts per proposal with no standardized templates.

Bottlenecks: 3-step auditing process is non-delegable and directly inflates proposal timelines. Multi-round reviews can take up to 15-20 days.

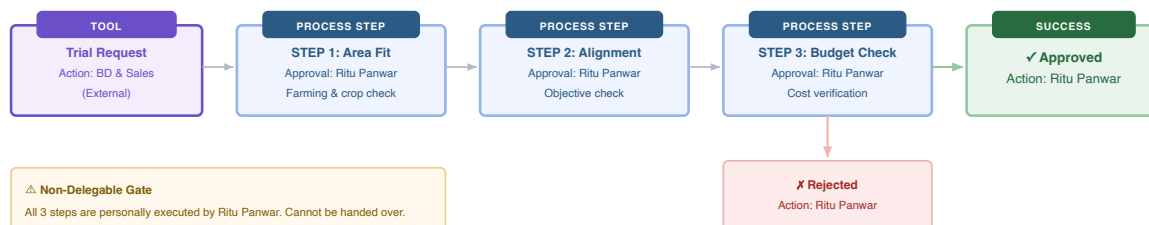
Dependencies: BD and Sales team are entirely reliant on Ritu Panwar for technical pitching materials and final sign-offs.



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Protocol Approval Gate — P-01 P-04



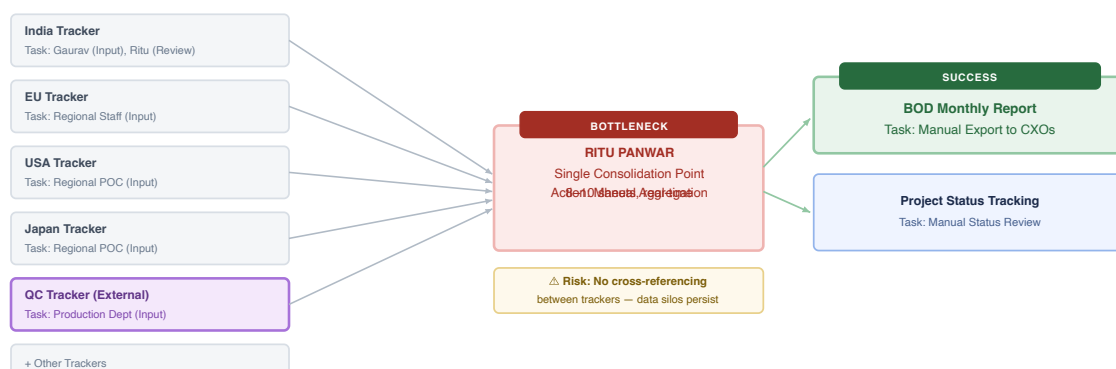
What is good: Gate system is well-defined and ensures rigorous technical assessment of trial requests.

What needs to be fixed: The compounded approval load of 18-20 active projects in peak season is entirely centralized on one person.

Bottlenecks: Non-delegable gate. All 3 steps are personally executed by the PD Manager, causing significant delays.

Dependencies: BD and Sales teams depend entirely on Ritu's bandwidth to move any trial forward.

Multi-Tracker Data Flow & Consolidation — P-02 P-03



What is good: Department maintains detailed, real-time trackers for every region and function.

What needs to be fixed: Data is siloed across 8-10 disconnected Google Sheets. There is no automated aggregation or cross-referencing between trackers.

Bottlenecks: Ritu is the sole consolidation point. The manual aggregation of these sheets delays project portfolio visibility.

Dependencies: R&D is dependent on timely updates from regional staff and Production/QC to keep trackers accurate.

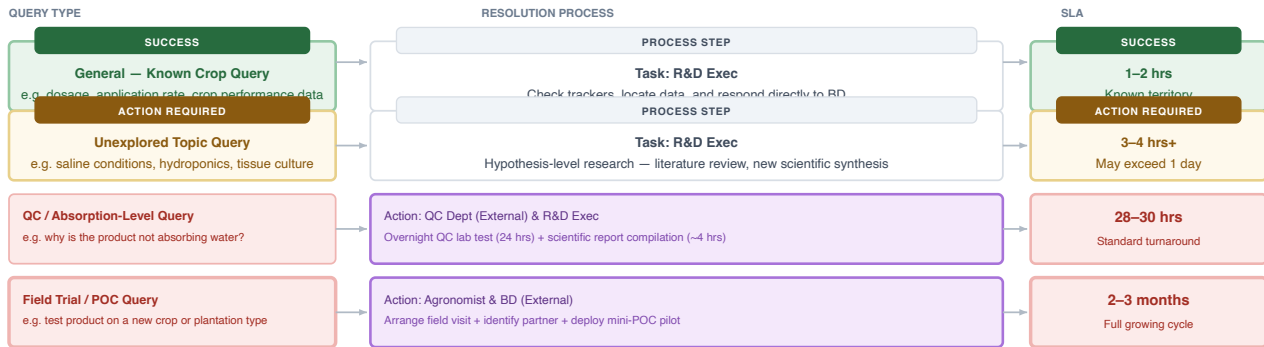


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Sales & BD → R&D Query Resolution — P-07



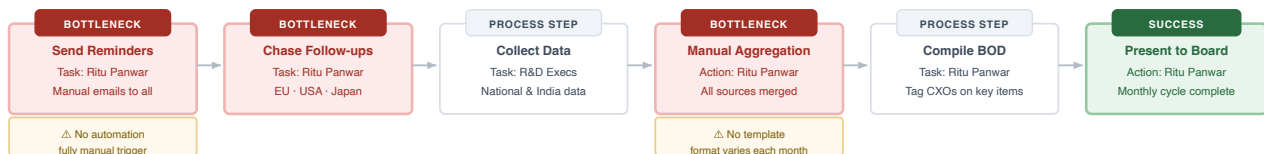
What is good: Query triage is naturally tiered by complexity and required effort.

What needs to be fixed: No self-serve knowledge base for Sales or BD. All query tiers are routed directly to R&D team members, disrupting focus time.

Bottlenecks: Critical-tier queries (QC-level and Field POC) have response times measured in days to months, stalling Sales closures.

Dependencies: Sales is entirely dependent on R&D for even basic known-crop data retrieval.

Monthly Reporting & BOD Aggregation — P-08



What is good: Consolidated monthly updates provide full visibility to the Board and CXOs.

What needs to be fixed: The reporting cycle runs manually with no automated reminders, no standardized aggregation template, and no pre-built BOD format.

Bottlenecks: Ritu personally coordinates across at least 3 time zones to collect data before she can begin compiling.

Dependencies: Aggregation depends completely on agronomists sending data on time; delays cascade down the line.



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SECTION 04 — CROSS-DEPARTMENT DEPENDENCIES

Cross-Department Dependencies

Findings and recommendations that require action outside the R&D Office to resolve

Sales & Business Development

COMMERCIAL SUPPORT

OPERATIONS DEPENDENCY

Sales heavily depends on R&D for technical inputs on B2B proposals and grants. R&D needs structured requests from Sales.

 **Key Person:** Sales Head / BD Lead

CRITICAL BOTTLENECK

Ad-hoc chat queries and unstructured proposal requests from Sales currently disrupt R&D's core research focus.

DATA TRANSFER & UPDATES NEEDED

Sales must adopt a standardized intake format for B2B requests. R&D must continuously push updated technical data to the SSOT Chatbot to enable Sales self-service.

Production & Quality Control (QC)

OPERATIONAL HANDOFF

OPERATIONS DEPENDENCY

R&D trial validation requires a strict 3-department handoff (Production lot → QC 24-hr test → R&D log). Furthermore, R&D relies on Production for raw material procurement (large-scale trials), machinery procurement advice, and joint-approvals for registrations/certifications.

 **Key Person:** Ritu Panwar (PD Manager)

CRITICAL BOTTLENECK

No established workflow exists for cross-departmental requests regarding raw materials, machinery, or certifications, causing breaks in the handoff chain and lost data.

DATA TRANSFER & UPDATES NEEDED

New workflows must be built into the dashboard to send pre-defined approval messages for raw materials and machinery. Live status updates must be synced from Production/QC systems to track testing statuses in real-time.



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Finance, HR & CXO Visibility

ADMIN & BUDGET SUPPORT

OPERATIONS DEPENDENCY

R&D teams (India & International) require rapid budget approvals from Finance for operational fieldwork. CXOs require top-down visibility into where these investments are going.

 **Key Persons: Shobha Shekhawat (Finance) / HR Manager**

CRITICAL BOTTLENECK

Delays in processing international payments stall fieldwork. Furthermore, CXOs lack immediate visibility into R&D's ROI and budget utilization.

DATA TRANSFER & UPDATES NEEDED

Budget requests and approvals must be directly integrated into the dashboard's visibility overview. This allows R&D Leads and CXOs to instantly track international budget allocations versus actual ROI generation. HR also requires updated KPIs for role standardization in Phase 2.



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SECTION 05 — OPERATIONAL PAIN POINTS

Pain Points Register

Ranked by operational impact. Each pain point is cross-referenced to its recommendation.

Priority Focus Areas:

- ① **Reporting & Visibility** — Real-time visibility of team member workloads, project timelines, and individual capacity is the most pressing gap. No system exists today to know who is doing what and when they will be free.
- ② **Trial Data Monitoring & Analysis** — No centralised tracking of ongoing trials, costs, outcomes, and statuses means the department cannot prevent overspend or learn from previous validation data.

	PAIN POINT	IMPACT	EVIDENCE FROM AUDIT
P-01	Severe Key-Person Dependency. All protocol approvals, grant writing, and international technical sign-offs are bottlenecked on a single person — the PD Manager (Ritu Panwar).	CRITICAL	3-stage approval process is non-delegable as currently structured; international proposals require up to 3 rounds and 15–20 days, stalling BD and Sales pipelines.
P-02	Scattered Trial & Project Data. Research, trial outcomes, and project status are fragmented across 8–10 disconnected Google Sheets with no cross-referencing or search.	CRITICAL	Causes redundant testing, data loss, and inability to retrieve past trial data — directly leading to preventable validation cost overruns.
P-03	No Trial Cost Monitoring or Spend Visibility. Validation is the largest cost center, but there is no centralised dashboard tracking trial costs, outcomes, and resource spend per project.	CRITICAL	Without visibility of previous trial data, the team risks repeating failed or inconclusive experiments — directly inflating the validation budget. Ritu confirmed repeat trials are a known cost driver.
P-04	No Project Management or Workload Tracking. PM tools (Asana) were abandoned; the team reverts to manual follow-up. No visibility of individual team member capacity or project timelines exists.	CRITICAL	Ritu confirmed she cannot know if a team member is overloaded before assigning new tasks. Cross-region data submissions are missed without automated reminders.



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P-05 Repetitive Agronomical Documentation Load. Back-to-back creation of national-side technical and agronomical reports is consuming a disproportionate share of Ritu Panwar's own time with no reusable template or delegation path.

HIGH

Ritu cited documentation as one of her two largest personal time sinks, specifically national-side technical insights and agronomical reports. These are written fresh each time with no standardised base template.

P-06 Proposal Drafting Fragmentation Across Members. B2B proposals (Arman), scientific proposals (Gaurav), and all international / grant proposals (Ritu) are siloed with no shared template library or intake format.

HIGH

Quarterly volume: 5–6 proposals in steady state, up to 20+ in a peak quarter. Each proposal requires a minimum of 2–3 drafts. No standardised format or proposal bank exists to accelerate repeat work.

P-07 Broken Information Flow to Sales & BD. Sales and BD teams regularly request technical data, crop dosage guidance, and pitch support but there is no self-serve retrieval mechanism — every query hits R&D directly.

HIGH

Simple queries take 1–2 hours; unexplored topics 3–4 hours; QC-level queries 28–30 hours standard turnaround. All routed manually through team members with no shared knowledge base for Sales and BD to self-serve.

P-08 Manual Reporting Reminders & BOD Aggregation. Monthly data collection from agronomists (including cross-timezone international contacts) is done via manual email reminders. Compiling the BOD report from multiple sources is entirely manual.

MEDIUM

Ritu confirmed she manually chases international agronomists for monthly data. The combined monthly report for CXOs, and the BOD-level summary, are assembled by hand each cycle with no automated aggregation.



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SECTION 06 — TIME & EFFORT

Time & Effort Allocation

Based on a standard 8-hour workday. Hours shown are estimates per team member per day, averaged across the team.

ACTIVITY	SHARE OF TEAM TIME	EST. TIME / DAY
Technical, Agronomical & Compliance Documentation	~30%	~2.4 hrs
Project Tracker Maintenance (Google Sheets)	~20%	~1.6 hrs
Grant & B2B Proposal Drafting	~20%	~1.6 hrs
New Product Acquisition & Partner Evaluation	~15%	~1.2 hrs
Protocol Auditing & Approvals	~10%	~48 min
Board & BOD Reporting Prep	~5%	~24 min



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SECTION 07 — CORRECTIVE ACTION PLAN

Recommended Solution Segment

This section outlines the integrated technology solution proposed to resolve operational bottlenecks, replacing the legacy recommendations register.

Implementation Required

Solution Modules to be Built:

1 R&D Frontend Dashboard

- **Real-Time Data Sync:** The dashboard links directly to your existing Google Sheets. If someone edits a Google Sheet, the dashboard updates instantly. If you make a change on the dashboard, it automatically saves back to the Google Sheet, meaning no data is ever out of sync.
- **Centralised Approval System:** A dedicated space where the R&D Head can instantly see all pending requests. You can request, review, approve, or reject trials and documents with a single click, eliminating delays caused by lost emails or messages.
- **Workload & Capacity Tracking:** A clear, visual breakdown of what every team member is currently working on. It helps the manager foresee delays, balance task distribution, and track exactly how much time is being spent on each project or trial.

2 AI & Knowledge Systems

- **Proposal Drafting Engine:** A custom AI assistant built specifically for your team that reads existing past proposals and templates securely stored on Google Drive. It automatically drafts new proposals in the exact required format, saving hours of manual writing. *Crucially, this system will live directly inside the new R&D dashboard so the team doesn't have to jump between multiple tools.*
- **Single Source of Truth (SSOT) Chatbot:** We will upgrade the existing chatbot so it has access to the entire R&D knowledge base. Sales and BD teams can ask it questions and get instant, accurate answers without interrupting the R&D team's actual work.

3 Automated Communications & Operations

- **Reporting Reminders:** Automated workflows via API (WhatsApp for internal, Slack for international, Gmail for external partners).
- **Unified OS integration:** Integrates into the larger EF Polymer operating system, giving CXOs role-based, one-click visibility.



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What Remains the Same

- **No New Backend Database:** Google Sheets and Drive remain the core sources.
- **Familiar Workflows:** The team can still update sheets manually without being forced to change underlying habits.

Tech Stack Utilized

- **Frontend:** ReactJS + TypeScript
- **APIs:** Google Sheets, Drive, WhatsApp, Slack, Gmail
- **AI:** Gemini/LLMs for RAG & Drafting
- **Host:** Google Cloud Platform

Key Operational Problems Solved

- Eliminates manual chasing of data and broken information flows across time zones.
- Resolves the massive documentation and proposal drafting bottleneck (saving significant PD Manager time).
- Centralizes tracking without forcing the team onto a new, complex PM tool.
- Provides CXOs and department heads instant visibility via a unified approval flow.

Future Scope of Improvement / Enhancement

- **AI Operations Agent:** In the future, an AI agent could be added to actively monitor task/trial data, identify bottlenecks, flag delays, and analyze team communications for proactive management. This is not needed right now but is a planned future enhancement.

Next Steps for Team Adoption

- **Mandatory Training:** Team adoption and training sessions must be made compulsory to ensure all team members understand how to use the dashboard and AI tools.
- **Feedback Loop:** Establish a continuous feedback loop to iterate on the solution on the ground, ensuring it genuinely adapts to the team's operational reality and achieves actual adoption.



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SECTION 08 — IMPLEMENTATION PLAN

Phased Roadmap

Initiatives sequenced across three phases, with quantified ROI and scope of work

INITIATIVE		PHASE 1 WEEKS 1–4	PHASE 2 WEEKS 5–9	PHASE 3 WEEKS 10–12
R&D Frontend Dashboard & Sync	Module 1	●		
AI Proposal Engine & Chatbot	Module 2		●	
Automated Comms & Unified OS	Module 3			●

Phase 1 — Core Infrastructure & Dashboard

Weeks 1–4 · Centralizing Data & Workflows

Maturity 1.5 → 2.5

SCOPE OF WORK & DEPENDENCIES

- **Build:** Connect existing Google Sheets to a new ReactJS Dashboard with Bi-Directional sync.
- **Build:** Implement the Approval Flow and Workload Management tracking modules.
- **Tech Stack & Badges:** React.js Dashboard Google Sheets API .
- **Team Adoption:** Conduct initial walkthrough sessions for the R&D team to navigate the new UI.
- **Client To-Do:** Provide read/write API access to the Google Workspace and map out the exact approval hierarchy.

100%

DATA SYNC

Eliminates manual copy-pasting across
disparate sheets.

Days Saved

CENTRALISED APPROVALS

Removes the international approval
bottleneck.

Live

CAPACITY TRACKING

Instant visibility into team workloads and
reversions.



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Phase 2 — AI & Knowledge Systems

Weeks 5–9 · Automating Documentation

Maturity 2.5 → 2.9

SCOPE OF WORK & DEPENDENCIES

- **Build:** Deploy custom Proposal Drafting Engine (via RAG over Google Drive) inside the dashboard.
- **Build:** Upgrade the existing RAG AI bot to serve as a centralized SSOT Chatbot, giving R&D, BD, Sales, and CXOs instant access to the full knowledge base.
- **Tech Stack & Badges:** Gemini LLMs Google Drive RAG .
- **Team Adoption:** Conduct prompt-engineering training and distribute formal adoption playbooks.
- **Client To-Do:** Organize Drive folders and provide raw historical proposals and templates to train the engine.

↓ 40-50%
DRAFTING TIME

Hours saved per proposal via GenAI automation.

Zero
SALES INTERRUPTIONS

Chatbot handles standard queries instantly.

Unified
TOOLING

AI tools live inside the dashboard; no app-jumping.

Phase 3 — Automated Comms & Operations

Weeks 10–12 · Scaling Visibility & Reminders

Maturity 2.9 → 3.2

SCOPE OF WORK & DEPENDENCIES

- **Build:** Integrate WhatsApp, Slack, and Gmail APIs for automated reporting and data collection reminders.
- **Build:** Connect the R&D dashboard into the broader EF Polymer OS for CXO-level access.
- **Tech Stack & Badges:** WhatsApp Business API Slack API Gmail API .
- **Team Adoption:** Establish a continuous feedback loop on the ground to iterate and sustain adoption.
- **Client To-Do:** Provide individual team member roadmaps, platform preferences, and contact details for API mapping.

Automated
DATA CHASING

Eliminates time zone delays via API reminders.

1-Click
CXO VISIBILITY

Instant board-level metrics via the Unified OS.

Sustained
USAGE

Feedback loop ensures the solution adapts to reality.



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SECTION 09 — RETURN ON INVESTMENT (ROI)

System Impact: Before & After

Measurable improvements post-implementation

CATEGORY	BEFORE (CURRENT STATE)	AFTER (POST-IMPLEMENTATION)
Data Tracking & Storage	8–10 scattered Google Sheets. High risk of data loss.	1 Centralised Dashboard. Real-time bi-directional sync (Single Source of Truth).
Approval Workflows	Manual WhatsApp/Slack messages, causing up to 15–20 day delays.	1-Click Digital Approvals. Notifications routed instantly, cutting delays by 80%.
Proposal & Grant Drafting	100% manual writing, consuming 20% of team bandwidth.	AI-Powered Drafting Engine. Generates structured drafts instantly, saving hours per proposal.
Sales/BD Inquiries	Ad-hoc interruptions to R&D for technical data.	SSOT Chatbot. Sales can self-serve technical answers securely without interrupting R&D.
Cross-Department Approvals	No established workflow for Production/QC handoffs.	Automated Status Sync. Live status updates tracking lot assignments and QC tests.
CXO & Financial Visibility	Budget tracking relies on manual reporting.	Executive Overview. Real-time dashboard visibility into ROI and budget allocation.
Department Maturity Score	1.5 / 5.0 (Ad-hoc)	3.2 / 5.0 (Systematised)



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Quantified Executive Time Savings

Estimated annual bandwidth reclaimed for the R&D Lead (Ritu Panwar & team) by automating proposal/grant drafting and eliminating sales query interruptions.

20% DRAFTING BANDWIDTH SAVED	80% APPROVAL DELAY CUT	22 hrs MONTHLY TIME SAVED	264 hrs ANNUAL HOURS SAVED
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Bandwidth Reinvestment Breakdown:

- **Current Friction:** R&D team spends ~20% of its working hours manually writing research proposals/grant decks and answering ad-hoc sales inquiries for technical data.
- **Automated Efficiency:** AI Proposal Drafting Engine and SSOT Knowledge Chatbot allow sales to self-serve technical queries and auto-generate proposal initial drafts.
- **Strategic Impact:** Reclaims **264 R&D hours per year (~33 full working days)** to focus on core polymer IP development, lab trial execution, and patent documentation.